

Session:13

1. Recursive function: reverse_string(s)

```
def reverse_string(s):  
    if len(s) <= 1:  
        return s  
    return s[-1] + reverse_string(s[:-1])
```

```
# Example  
print(reverse_string("hello")) # olleh
```

2. Recursive function: sum_playlist_durations(durations)

```
def sum_playlist_durations(durations):  
    if len(durations) == 0:  
        return 0  
    return durations[0] + sum_playlist_durations(durations[1:])
```

```
# Example  
print(sum_playlist_durations([200, 180, 240])) # 620
```

3. Local vs Global variable explanation

Code:

```
count = 10  
  
def update_count():  
    count = 5  
    print('Inside:', count)  
  
update_count()  
print('Outside:', count)
```

Explanation:

- `count = 10` → **global variable**
- Inside `update_count()`, `count = 5` → **local variable** (this does NOT modify the global one)

What gets printed:

Inside: 5
Outside: 10

Why?

- The function creates a **new local variable** `count`
 - The global `count` remains unchanged
-

4. Recursive function: `count_likes(posts)`

Assumption: structure like this:

```
posts = {
  "likes": 10,
  "replies": [
    {
      "likes": 5,
      "replies": [
        {"likes": 2, "replies": []}
      ]
    },
    {
      "likes": 3,
      "replies": []
    }
  ]
}
```

Solution:

```
def count_likes(post):
    total = post.get("likes", 0)

    for reply in post.get("replies", []):
        total += count_likes(reply)

    return total

# Example usage
post_data = {
    "likes": 10,
    "replies": [
        {
            "likes": 5,
            "replies": [
                {"likes": 2, "replies": []}
            ]
        },
        {
```

```

        "likes": 3,
        "replies": []
    }
]
}

print(count_likes(post_data)) # 20

```

5. Script: local vs global variable lifetime (WhatsApp-style status)

```

# Global variable (app-wide status)
app_status = "Online"

def update_status():
    # Local variable (only exists inside function)
    user_status = "Typing..."

    print("During function call:")
    print("User status (local):", user_status)
    print("App status (global):", app_status)

    return user_status

print("Before function call:")
print("App status:", app_status)

returned_status = update_status()

print("\nAfter function call:")
print("App status:", app_status)

# user_status is NOT accessible here (would cause error if uncommented)
# print(user_status)
print("Returned user status:", returned_status)

```